EXPLAINABLE HYBRID MACHINE LEARNING FRAMEWORK FOR PREDICTIVE MAINTENANCE

Section: 2 | Members: 2520030019 – V. Navadeep, 2520030063 – K. Shankar
Guide: Dr. N. Nirmalajyothi | Department of Computer Science and Engineering | KLH University | A.Y.: 2026-27

Abstract

This project presents an explainable hybrid machine learning framework for predictive maintenance using the AI4I 2020 dataset. The system analyzes machine sensor data to predict potential failures before they occur. A hybrid model combining Random Forest and XGBoost is used for better accuracy, and SHAP is applied to explain the predictions, providing transparent and trustworthy maintenance decisions.

Tools & Technologies

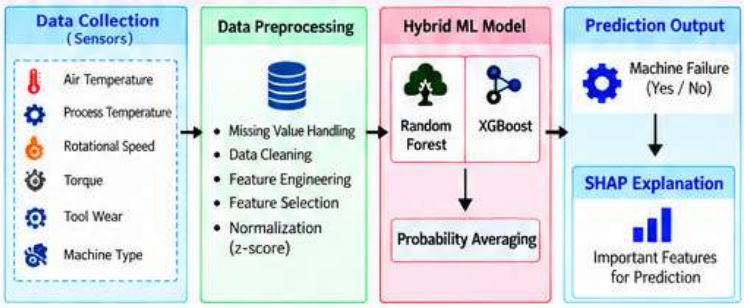
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook / Google Colab / VS Code
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Machine Learning:** Logistic Regression, Decision Tree, Random Forest, XGBoost (Hybrid)
- **Explainable AI:** SHAP
- **Dataset:** AI4I 2020 Predictive Maintenance Dataset (CSV)
- **Tools:** Git & GitHub

Problem Statement & Objectives

Problem: Unexpected machine failures in industries lead to production downtime, high maintenance costs and reduced efficiency. Traditional maintenance methods are either reactive or time-based and do not effectively predict actual failures.

- **Objective 1:** Predict potential machine failures using sensor data.
- **Objective 2:** Improve prediction accuracy using a hybrid model (Random Forest + XGBoost).
- **Objective 3:** Provide explainability using SHAP to identify important features.
- **Objective 4:** Help maintenance teams plan maintenance and reduce downtime, cost and equipment damage.
- **Scope:** Industrial machines, maintenance teams and plant operators.

System Architecture



Existing vs Proposed System

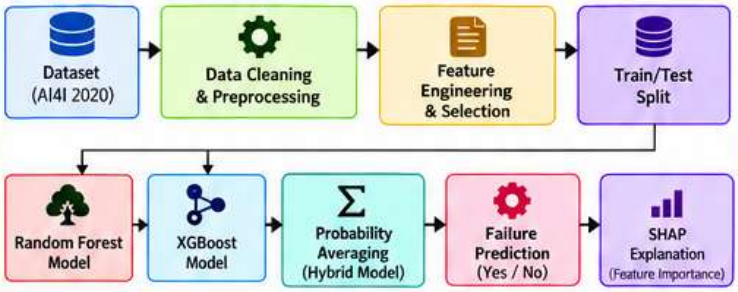
Feature	Existing System	Proposed System
Approach	Traditional / Single model	Hybrid ML (RF + XGBoost)
Prediction	Lower accuracy	Higher accuracy and reliable
Explainability	Not available	SHAP-based feature explanation
Data Handling	Limited preprocessing	Comprehensive preprocessing and feature engineering
Maintenance Planning	Reactive / Time-based	Predictive and data-driven
Usage	Less trust and transparency	More interpretable and actionable insights

Dataset Information

- **Dataset:** AI4I 2020 Prodictive Maintenance Dataset
- **Total Records:** 10,000
- **Features:** Air temperature, Process temperature, Rotational speed, Torque, Tool wear, Machine type
- **Target Variable:** Machine Failure (0: No, 1: Yes)
- **Format:** CSV
- **Description:** Sensor data collected from industrial machines with operating parameters.

Feature	Type
Air temperature	Numerical
Process temperature	Numerical
Rotational speed	Numerical
Torque	Numerical
Tool wear	Numerical
Machine type	Categorical
Machine failure	Binary (Target)

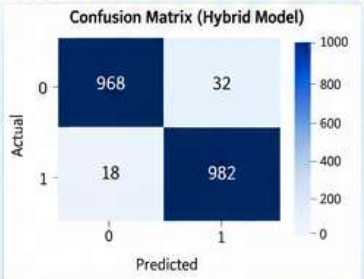
Proposed System Flow Diagram

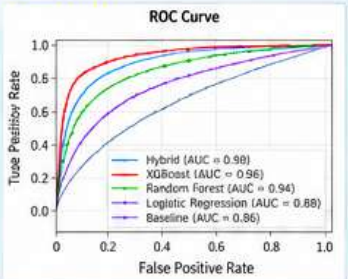


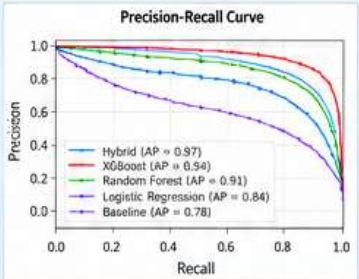
Results & Model Performance

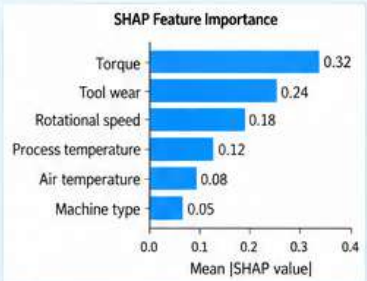
Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Existing Study (literature baseline)	91.2%	78.4%	70.1%	74.0%	0.86
Logistic Regression	92.5%	80.2%	73.6%	76.8%	0.88
Decision Tree	94.1%	84.3%	81.0%	82.6%	0.92
Random Forest	96.1%	88.7%	85.3%	86.9%	0.94
XGBoost (standalone)	97.3%	91.5%	89.2%	90.3%	0.96
Proposed Hybrid (RF + XGBoost)	**98.4%**	**94.8%**	**93.6%**	**94.2%**	**0.98**

Visual Results and Discussion









Conclusion

This project developed an explainable hybrid machine learning framework for predictive maintenance using the AI4I 2020 dataset. The hybrid model (Random Forest + XGBoost) achieved high accuracy, and SHAP provided feature-level explanations, making the system more interpretable and useful for real-world predictive-maintenance applications.

Future Scope

- Apply the model to real-time IIoT sensor data.
- Integrate with an industrial dashboard for live monitoring.
- Predict specific fault types instead of only binary failure.
- Use deep learning models (e.g., LSTM) for time-series data.
- Extend the system to multiple machine types and larger datasets.